

Program No: 1	Date: 29/06/2026
Program Title: Create an app that displays name and email id.	
MainActivity.java:	
<pre> package com.example.labcycle; import android.annotation.SuppressLint; import android.os.Bundle; import androidx.activity.EdgeToEdge; import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets; import androidx.core.view.ViewCompat; import androidx.core.view.WindowInsetsCompat; import android.view.View; import android.widget.EditText; import android.widget.Button; import android.widget.TextView; public class MainActivity extends AppCompatActivity { EditText name,email; Button btn; TextView t1,t2; @SuppressWarnings("MissingInflatedId") @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); EdgeToEdge.enable(this); setContentView(R.layout.activity_main); name = findViewById(R.id.nameetxt); email = findViewById(R.id.emailTxt); btn = findViewById(R.id.btnclick); t1 = findViewById(R.id.txtView1); t2 = findViewById(R.id.txtView2); btn.setOnClickListener(new View.OnClickListener() { @Override public void onClick(View view) { String n1 = name.getText().toString(); String n2 = email.getText().toString(); t1.setText(n1); t2.setText(n2); } }); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout </pre>	

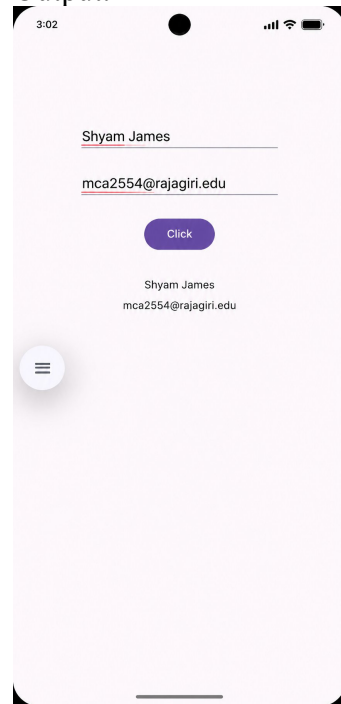
```

xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="match_parent">
<EditText
    android:id="@+id/nametxt"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Name"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="150dp"/>
<EditText
    android:id="@+id/emailTxt"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Email"
    app:layout_constraintTop_toBottomOf="@id/nametxt"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="16dp"/>
<Button
    android:id="@+id/btnclick"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click"
    app:layout_constraintTop_toBottomOf="@id/emailTxt"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="20dp"/>
<TextView
    android:id="@+id/txtView1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Name"
    app:layout_constraintTop_toBottomOf="@id/btnclick"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="30dp"/>
<TextView
    android:id="@+id/txtView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Email"
    app:layout_constraintTop_toBottomOf="@id/txtView1"

```

```
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintEnd_toEndOf="parent"  
android:layout_marginTop="10dp"/>  
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 2	Date: 29/06/2026
Program Title: Create an app that takes marks of any three subjects from the user and displays its sum and average.	
MainActivity.java:	
<pre> package com.example.labcycle; import android.annotation.SuppressLint; import android.os.Bundle; import androidx.activity.EdgeToEdge; import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets; import androidx.core.view.ViewCompat; import androidx.core.view.WindowInsetsCompat; import android.view.View; import android.widget.EditText; import android.widget.Button; import android.widget.TextView; public class MainActivity extends AppCompatActivity { EditText m1, m2, m3; Button btn; TextView t1, t2; @SuppressWarnings("MissingInflatedId") @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); EdgeToEdge.enable(this); setContentView(R.layout.activity_main); m1 = findViewById(R.id.mark1); m2 = findViewById(R.id.mark2); m3 = findViewById(R.id.mark3); btn = findViewById(R.id.btncal); t1 = findViewById(R.id.textView4); t2 = findViewById(R.id.textView5); btn.setOnClickListener(new View.OnClickListener() { @Override public void onClick(View view) { int n1 = Integer.parseInt(m1.getText().toString()); int n2 = Integer.parseInt(m2.getText().toString()); int n3 = Integer.parseInt(m3.getText().toString()); int sums = n1 + n2 + n3; int avg = sums / 3; t1.setText("Sum is " + sums); } }); } } </pre>	


```

        t2.setText("Average is " + avg);
    }
    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/mark1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="92dp"
        android:layout_marginBottom="43dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark1"
        app:layout_constraintBottom_toTopOf="@+id/mark2"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText
        android:id="@+id/mark2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="91dp"
        android:layout_marginBottom="40dp"
        android:ems="10"
        android:inputType="number"
        android:text="mark2"
        app:layout_constraintBottom_toTopOf="@+id/mark3"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText
        android:id="@+id/mark3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="85dp"
        android:layout_marginBottom="48dp"
        android:ems="10"

```

```

        android:inputType="numberSigned"
        android:text="mark3"
        app:layout_constraintBottom_toTopOf="@+id/btnCal"
        app:layout_constraintEnd_toEndOf="parent" />

<Button
    android:id="@+id/btnCal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="182dp"
    android:layout_marginBottom="49dp"
    android:text="Button"
    app:layout_constraintBottom_toTopOf="@+id/textView4"
    app:layout_constraintEnd_toEndOf="parent" />

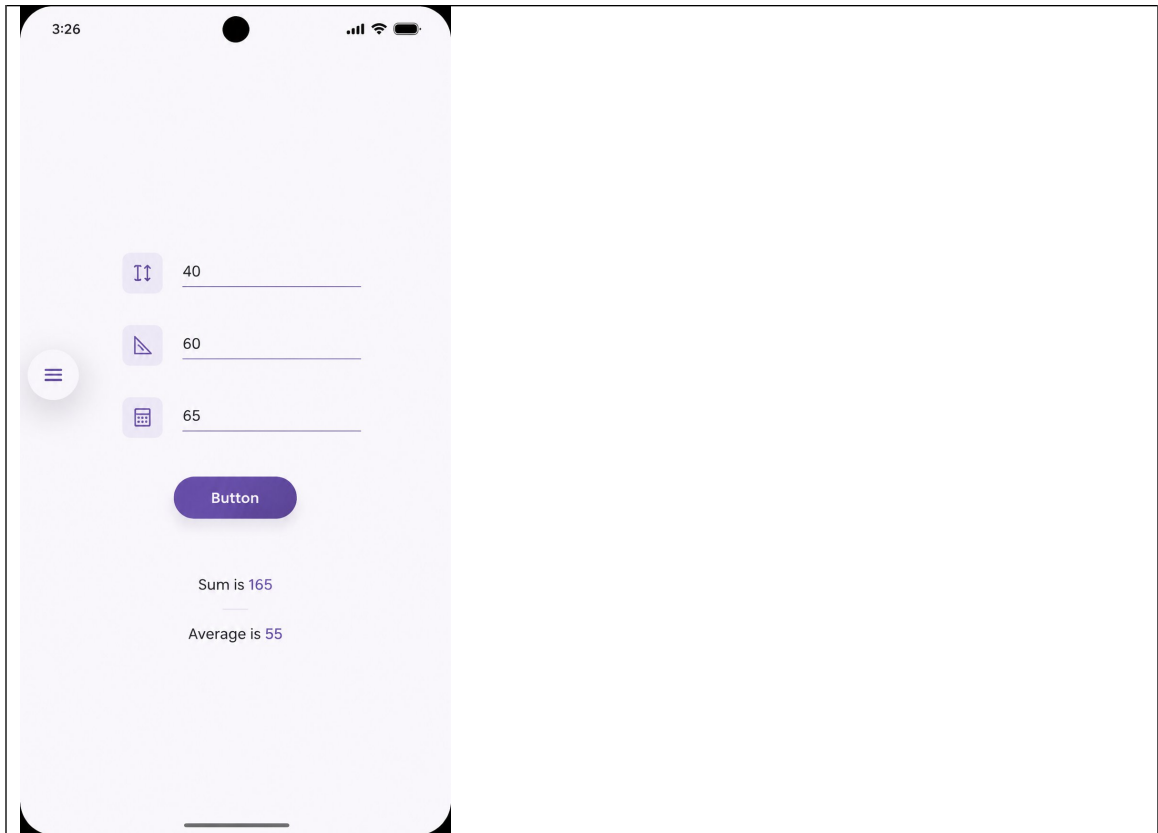
<TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="202dp"
    android:layout_marginBottom="160dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="200dp"
    android:layout_marginBottom="120dp"
    android:text="TextView"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 3	Date: 29/06/2026
Program Title: Create suitable apps to design the different layouts of the androidstudio.	
MainActivity.java:	
<pre> package com.example.labcycle; import android.os.Bundle; import android.widget.Button; import android.widget.Toast; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { Button button1, button2, button3; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); button1 = findViewById(R.id.button1); button2 = findViewById(R.id.button2); button3 = findViewById(R.id.button3); button1.setOnClickListener(v -> Toast.makeText(this, "LinearLayout Button 1", Toast.LENGTH_SHORT).show()); button2.setOnClickListener(v -> Toast.makeText(this, "LinearLayout Button 2", Toast.LENGTH_SHORT).show()); button3.setOnClickListener(v -> Toast.makeText(this, "RelativeLayout Button", Toast.LENGTH_SHORT).show()); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <ScrollView xmlns:android="http://schemas.android.com/apk/res/android" android:layout_width="match_parent" android:layout_height="match_parent"> <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="vertical" android:padding="16dp"> </pre>	

```

<!-- LinearLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Linear Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="20dp">

    <Button
        android:id="@+id/button1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 1"/>

    <Button
        android:id="@+id/button2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Button 2"/>
</LinearLayout>

<!-- RelativeLayout -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Relative Layout"
    android:textStyle="bold"
    android:textSize="18sp"/>

<RelativeLayout
    android:layout_width="match_parent"
    android:layout_height="120dp"
    android:background="#DDDDDD"
    android:layout_marginBottom="20dp">

    <TextView
        android:id="@+id/text1"
        android:layout_width="wrap_content"

```

```
        android:layout_height="wrap_content"
        android:text="Top Left"/>

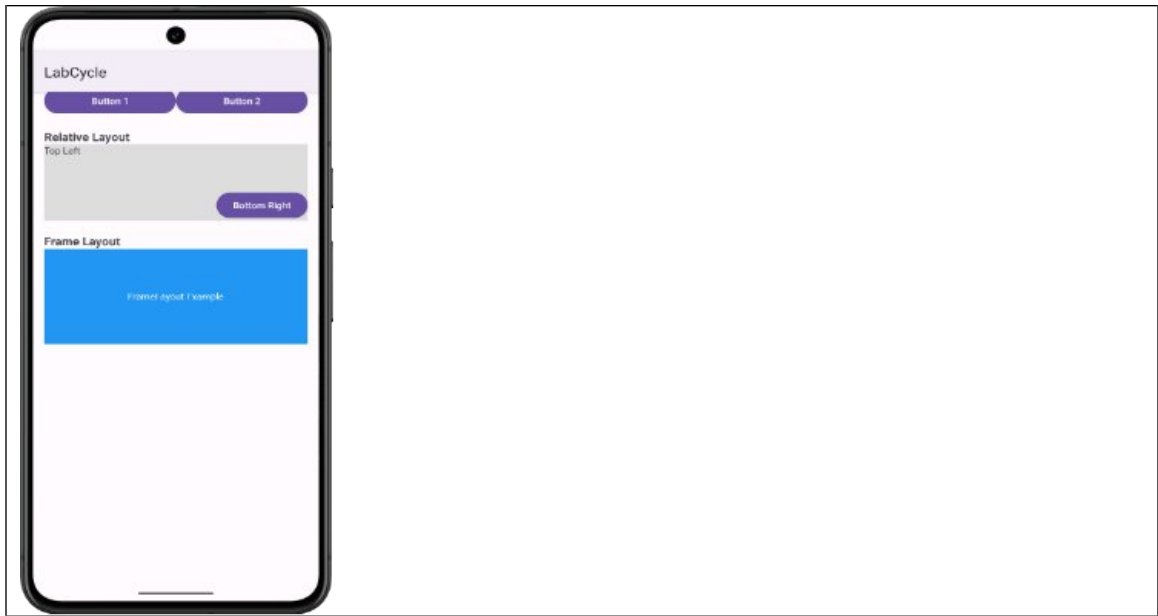
        <Button
            android:id="@+id/button3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Bottom Right"
            android:layout_alignParentBottom="true"
            android:layout_alignParentEnd="true"/>
    </RelativeLayout>

    <!-- FrameLayout -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Frame Layout"
        android:textStyle="bold"
        android:textSize="18sp"/>

    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:background="#2196F3">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="FrameLayout Example"
            android:textColor="#FFFFFF"
            android:layout_gravity="center"/>
    </FrameLayout>
</LinearLayout>
</ScrollView>
```

Output:



Program No: 4	Date: 29/06/2026
Program Title: Develop a simple calculator app.	
MainActivity.java:	
<pre> package com.example.pg04_calculator; import android.os.Bundle; import androidx.activity.EdgeToEdge; import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets; import androidx.core.view.ViewCompat; import androidx.core.view.WindowInsetsCompat; import android.widget.TextView; import android.widget.Button; import android.widget.EditText; public class MainActivity extends AppCompatActivity { TextView result; Button ad, sub, div, mult; EditText no1, no2; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); EdgeToEdge.enable(this); setContentView(R.layout.activity_main); result = findViewById(R.id.result); ad = findViewById(R.id.ad); sub = findViewById(R.id.sub); div = findViewById(R.id.div); mult = findViewById(R.id.mult); no1 = findViewById(R.id.no1); no2 = findViewById(R.id.no2); ad.setOnClickListener(v -> { int a = Integer.parseInt(no1.getText().toString()); int b = Integer.parseInt(no2.getText().toString()); int res = a + b; result.setText(res+""); }); sub.setOnClickListener(v -> { int a = Integer.parseInt(no1.getText().toString()); int b = Integer.parseInt(no2.getText().toString()); int res = a - b; result.setText(res+""); }); </pre>	


```

mult.setOnClickListener(v -> {
    int a = Integer.parseInt(no1.getText().toString());
    int b = Integer.parseInt(no2.getText().toString());
    int res = a * b;
    result.setText(res+"");

});

div.setOnClickListener(v -> {
    int a = Integer.parseInt(no1.getText().toString());
    int b = Integer.parseInt(no2.getText().toString());
    if(b!=0) {
        int res = a / b;
        result.setText(res+"");
    }
    else
        result.setText("Error: Divsion / 0");
});
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Result: "
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.252"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<TextView
    android:id="@+id/result"

```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.52"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

<EditText
    android:id="@+id/no1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter First Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.228" />

<EditText
    android:id="@+id/no2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Enter Second Number"
    android:inputType="text"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.437"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.355" />

<Button
    android:id="@+id/ad"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Add"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"

```

```
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.655" />
```

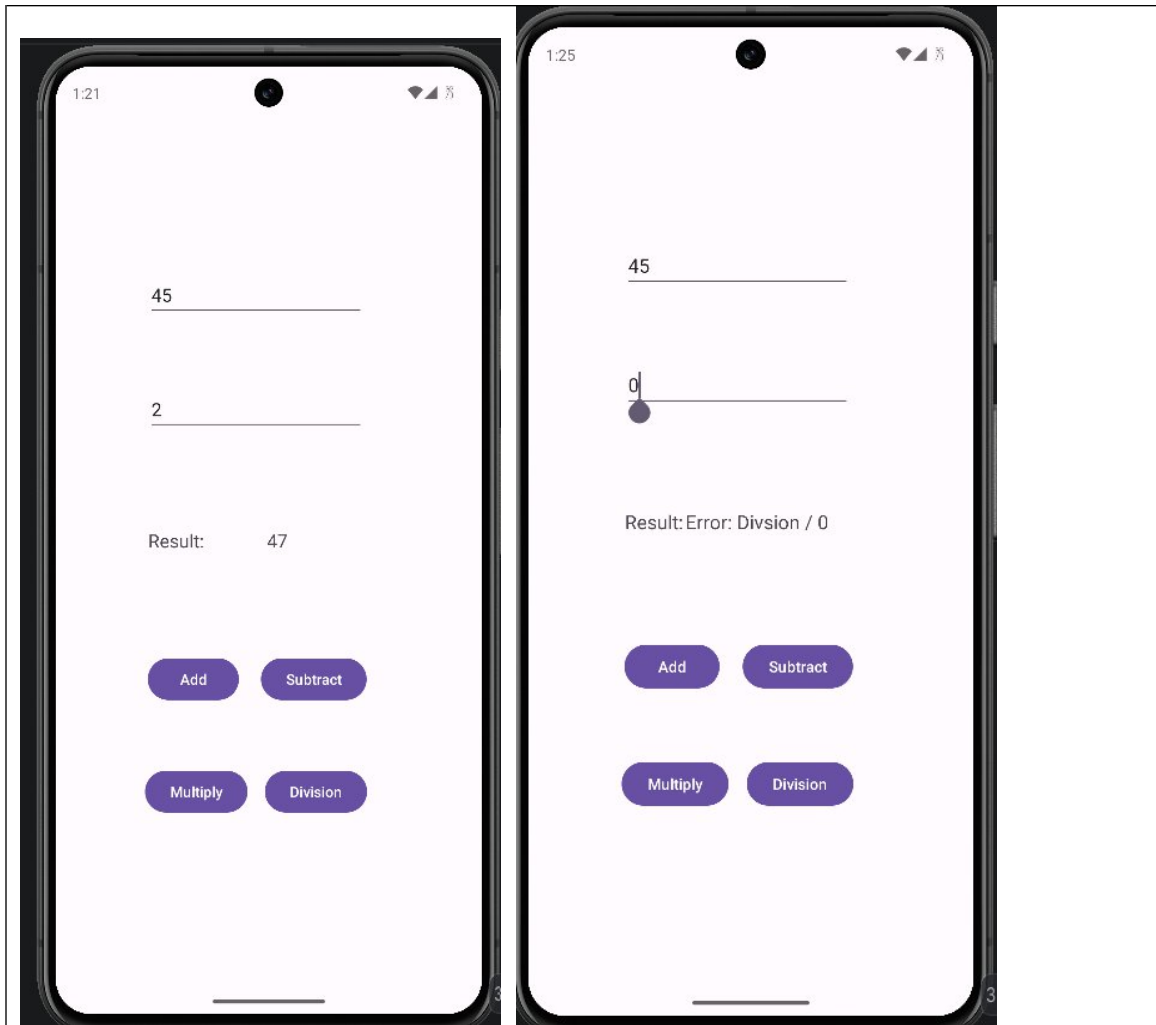
```
<Button
    android:id="@+id/sub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Subtract"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.64"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.655" />
```

```
<Button
    android:id="@+id/mult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Multiply"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.275"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.78" />
```

```
<Button
    android:id="@+id/div"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Division"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.646"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.78" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



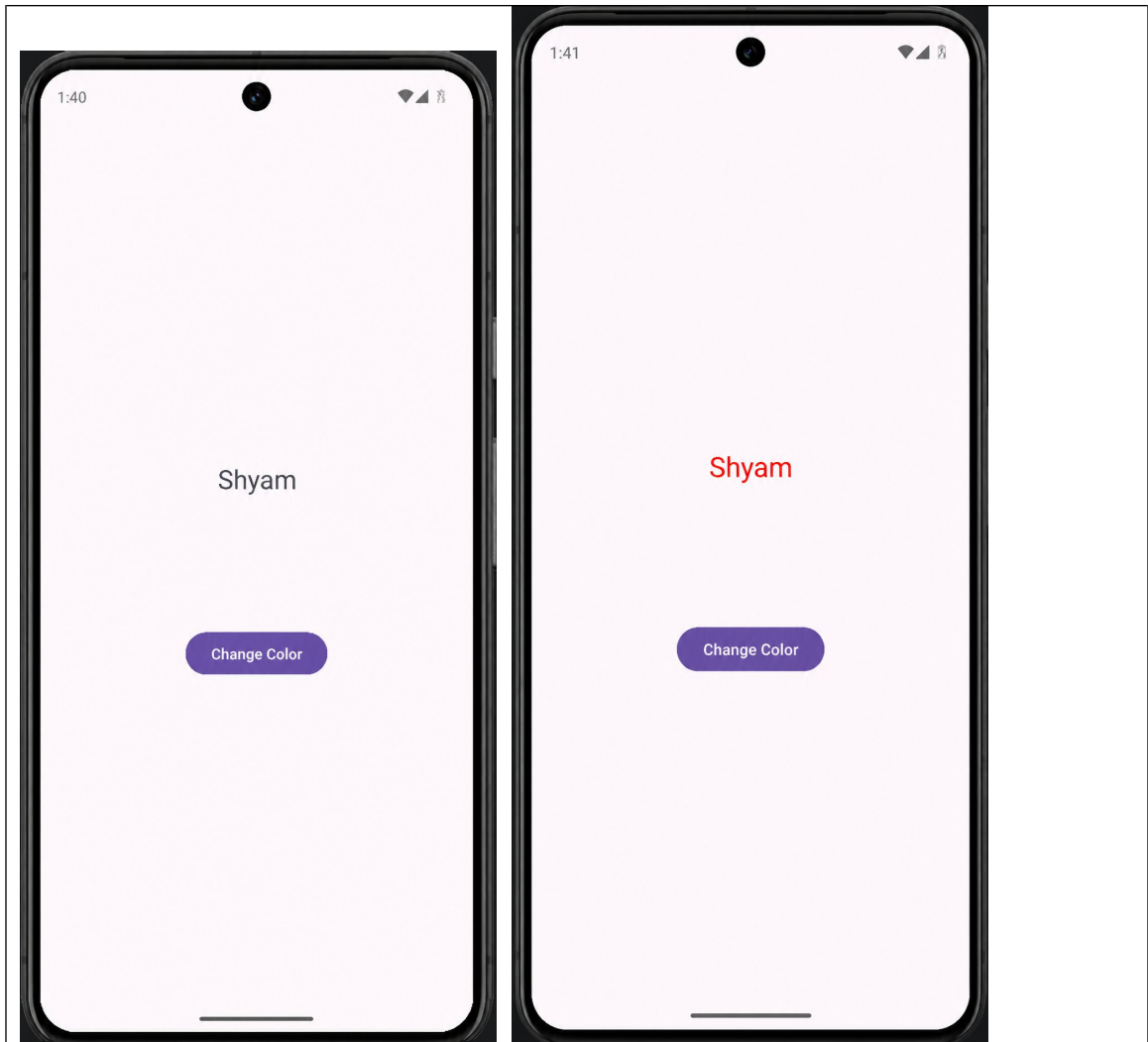
Program No: 5	Date: 29/06/2026
Program Title: Create an app that changes text color when a button is clicked.	
MainActivity.java:	
<pre> package com.example.pg05_textcolor; import android.graphics.Color; import android.os.Bundle; import androidx.activity.EdgeToEdge; import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets; import androidx.core.view.ViewCompat; import androidx.core.view.WindowInsetsCompat; import android.widget.TextView; import android.widget.Button; public class MainActivity extends AppCompatActivity { TextView name; Button color; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); EdgeToEdge.enable(this); setContentView(R.layout.activity_main); color = findViewById(R.id.color); name = findViewById(R.id.name); color.setOnClickListener(v -> { name.setTextColor(Color.RED); }); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main" android:layout_width="match_parent" android:layout_height="match_parent" tools:context=".MainActivity"> <TextView android:id="@+id/name" </pre>	

```
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Shyam"
android:textSize="24sp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.498"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.421" />
```

```
<Button
    android:id="@+id/color"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Change Color"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.613" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 6	Date: 29/06/2026
Program Title: Create an app that displays Date and Time in different formats using Linear Layout.	
MainActivity.java:	
<pre> package com.example.pg06_datetime; import android.os.Bundle; import android.view.View; import android.widget.Button; import android.widget.TextView; import androidx.appcompat.app.AppCompatActivity; import java.text.SimpleDateFormat; import java.util.Date; import java.util.Locale; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); final TextView textView1 = findViewById(R.id.textView1); final TextView textView2 = findViewById(R.id.textView2); final TextView textView3 = findViewById(R.id.textView3); final TextView textView4 = findViewById(R.id.textView4); Button button1 = findViewById(R.id.button1); button1.setOnClickListener(new View.OnClickListener() { @Override public void onClick(View v) { Date currentDateTime = new Date(); SimpleDateFormat format1 = new SimpleDateFormat("EEEE, dd MMM yyyy", Locale.getDefault()); SimpleDateFormat format2 = new SimpleDateFormat("MM/dd/yyyy", Locale.getDefault()); SimpleDateFormat format3 = new SimpleDateFormat("HH:mm:ss z", Locale.getDefault()); SimpleDateFormat format4 = new SimpleDateFormat("hh:mm a", Locale.getDefault()); textView1.setText(format1.format(currentDateTime)); textView2.setText(format2.format(currentDateTime)); textView3.setText(format3.format(currentDateTime)); textView4.setText(format4.format(currentDateTime)); } }); } } </pre>	

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

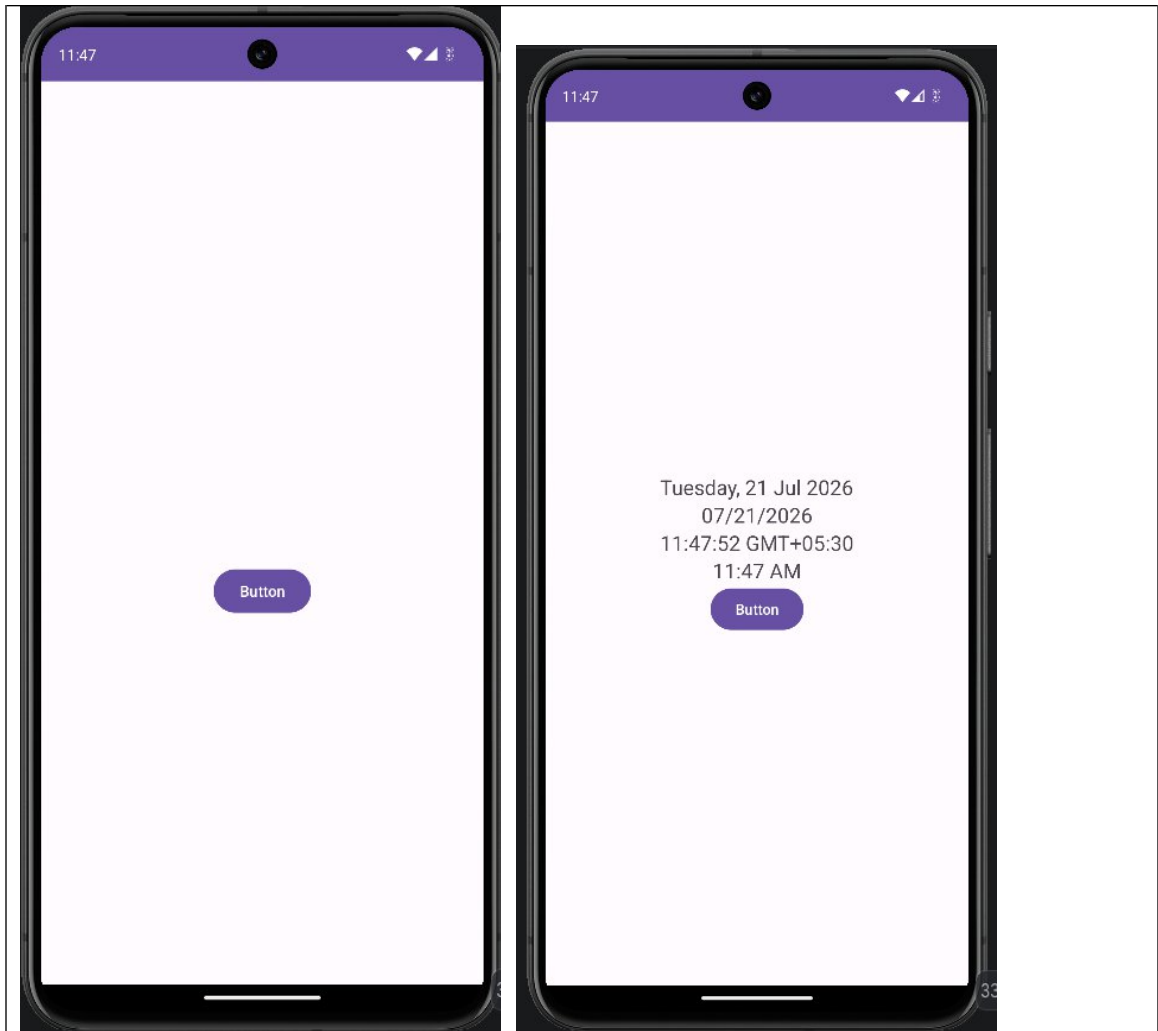
    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button" />

</LinearLayout>
```

Output:



Program No: 7	Date: 29/06/2026
Program Title: Create an app that displays a Toast on the launch of an activity.	
MainActivity.java:	
<pre> package com.example.pg07_launchtoast; import android.os.Bundle; import android.widget.Toast; import androidx.activity.EdgeToEdge; import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets; import androidx.core.view.ViewCompat; import androidx.core.view.WindowInsetsCompat; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); EdgeToEdge.enable(this); setContentView(R.layout.activity_main); Toast.makeText(this, "Welcome Shyam", Toast.LENGTH_SHORT).show(); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main" android:layout_width="match_parent" android:layout_height="match_parent" tools:context=".MainActivity"> <TextView android:layout_width="wrap_content" android:layout_height="wrap_content" android:text="Hello ^_^" android:textSize="24sp" app:layout_constraintBottom_toBottomOf="parent" app:layout_constraintEnd_toEndOf="parent" app:layout_constraintStart_toStartOf="parent" app:layout_constraintTop_toTopOf="parent" /> </androidx.constraintlayout.widget.ConstraintLayout> </pre>	

Output:



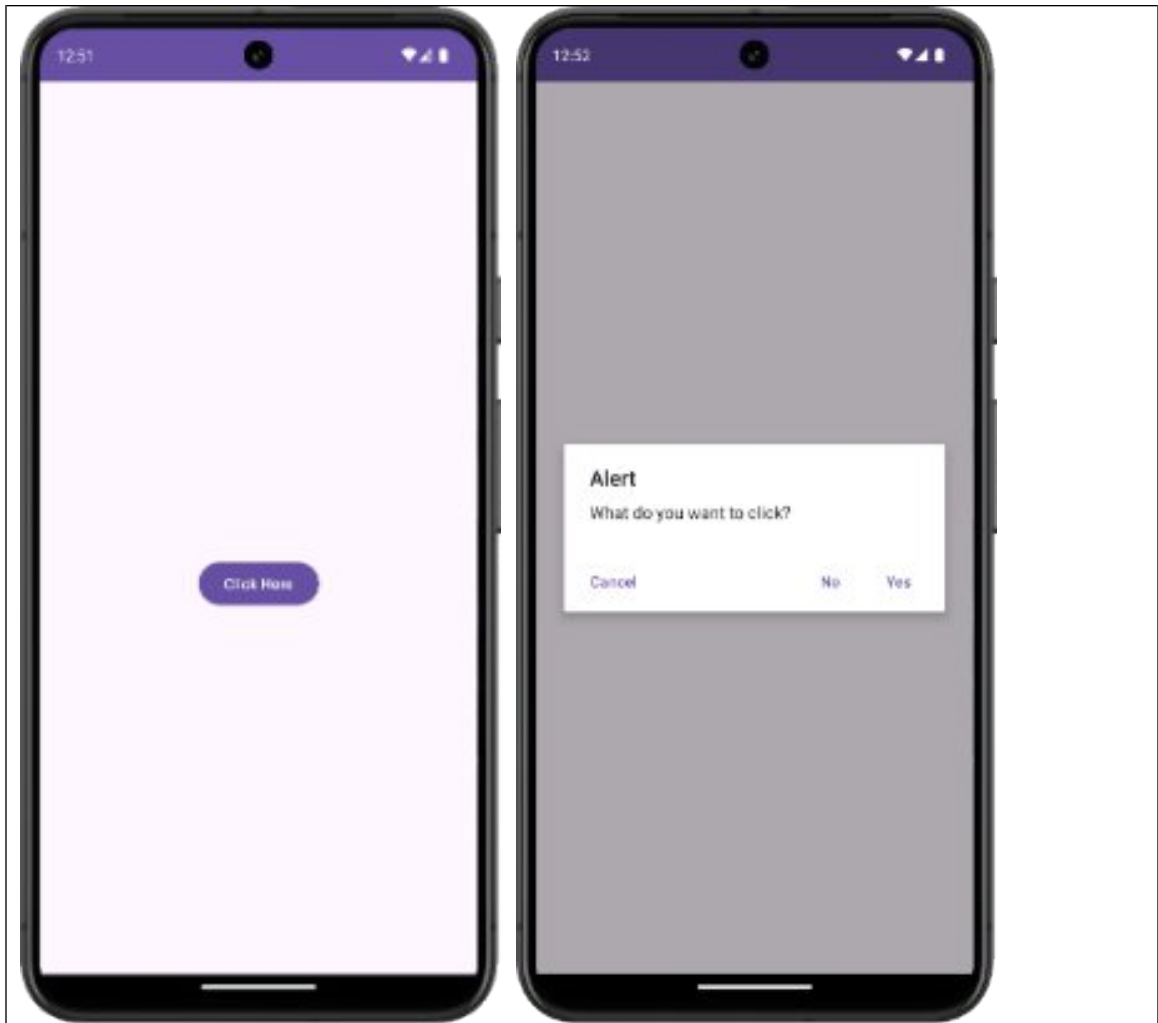
Program No: 8	Date: 29/06/2026
Program Title: Create an app that creates an Alert Dialog Box.	
MainActivity.java:	
<pre> package com.example.pg08_alertbox; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import androidx.appcompat.app.AlertDialog; import android.widget.Button; import android.widget.Toast; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); Button btn = findViewById(R.id.btn); btn.setOnClickListener(v -> { AlertDialog.Builder builder = new AlertDialog.Builder(MainActivity.this); builder.setTitle("Alert"); builder.setMessage("What do you want to click?"); builder.setPositiveButton("Yes", (dialog, which) -> { Toast.makeText(MainActivity.this, "You clicked Yes", Toast.LENGTH_SHORT).show(); }); builder.setNegativeButton("No", (dialog, which) -> { Toast.makeText(MainActivity.this, "You clicked No", Toast.LENGTH_SHORT).show(); }); builder.setNeutralButton("Cancel", (dialog, which) -> { dialog.dismiss(); }); builder.show(); }); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" </pre>	

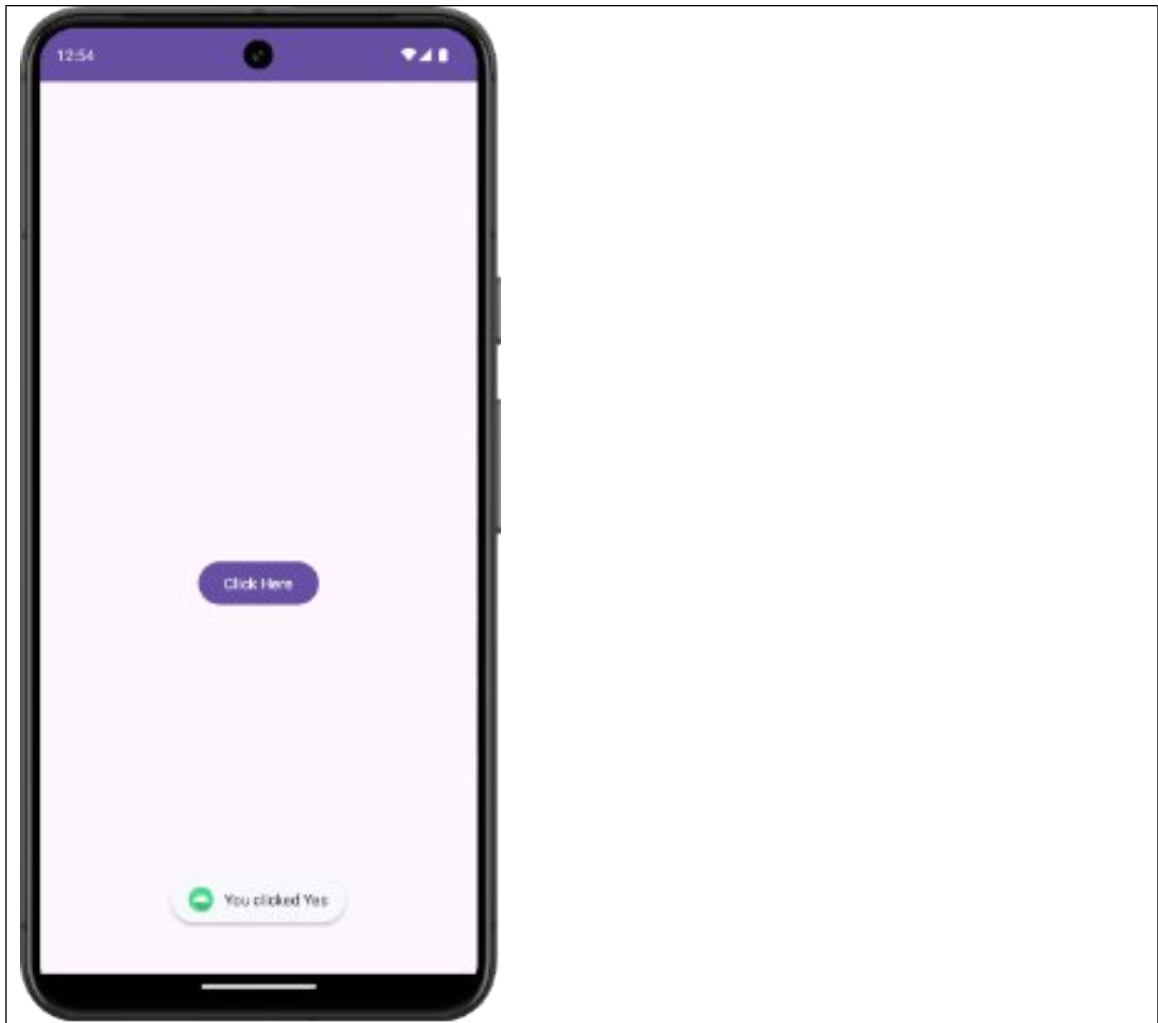
```
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
```

```
<Button
    android:id="@+id/btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click Here"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.566" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:





Program No: 9	Date: 29/06/2026
Program Title: Create a Greeting card app that displays an image.	
MainActivity.java:	
<pre>package com.example.pg09_greetingcard; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); } }</pre>	
activity_main.xml:	
<pre><?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main" android:layout_width="match_parent" android:layout_height="match_parent" tools:context=".MainActivity"> <ImageView android:id="@+id/imageView" android:layout_width="288dp" android:layout_height="393dp" app:layout_constraintBottom_toBottomOf="parent" app:layout_constraintEnd_toEndOf="parent" app:layout_constraintHorizontal_bias="0.495" app:layout_constraintStart_toStartOf="parent" app:layout_constraintTop_toTopOf="parent" app:layout_constraintVertical_bias="0.328" app:srcCompat="@drawable/greetings" /> <TextView android:id="@+id/textView2" android:layout_width="wrap_content" android:layout_height="wrap_content" android:text="Happy Birthday!" android:textColor="#2196F3" android:textSize="24sp"</pre>	

```
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.739" />

<TextView
    android:id="@+id/textView"
    android:layout_width="258dp"
    android:layout_height="59dp"
    android:text="Wishing you a joyful day and an amazing year ahead!"
    android:textAlignment="center"
    android:textColor="#020202"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.496"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.833" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 10	Date: 29/06/2026
Program Title: Create an app that plays audio when we click on the button ``Play Music" and plays video when we click on the button ``Play Video".	
MainActivity.java:	
<pre> package com.example.pg10_videomusic; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.widget.Button; import android.widget.VideoView; import android.net.Uri; import android.media.MediaPlayer; import android.widget.MediaController; public class MainActivity extends AppCompatActivity { MediaPlayer mediaPlayer; VideoView videoView; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); Button btnMusic = findViewById(R.id.music); Button btnVideo = findViewById(R.id.video); videoView = findViewById(R.id.videoView); videoView.setZOrderOnTop(true); mediaPlayer = MediaPlayer.create(this, R.raw.song); btnMusic.setOnClickListener(v -> { if (!mediaPlayer.isPlaying()) { mediaPlayer.start(); } }); btnVideo.setOnClickListener(v -> { Uri uri = Uri.parse("android.resource://" + getPackageName() + "/" + R.raw.video); videoView.setVideoURI(uri); MediaController controller = new MediaController(MainActivity.this); videoView.setMediaController(controller); controller.setAnchorView(videoView); videoView.start(); }); } } </pre>	

```

    });
}

@Override
protected void onDestroy() {
    super.onDestroy();
    if (mediaPlayer != null) {
        mediaPlayer.release();
        mediaPlayer = null;
    }
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <VideoView
        android:id="@+id/videoView"
        android:layout_width="348dp"
        android:layout_height="231dp"
        android:background="#000000"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.492"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.23" />

    <Button
        android:id="@+id/music"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play Music"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.237"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"

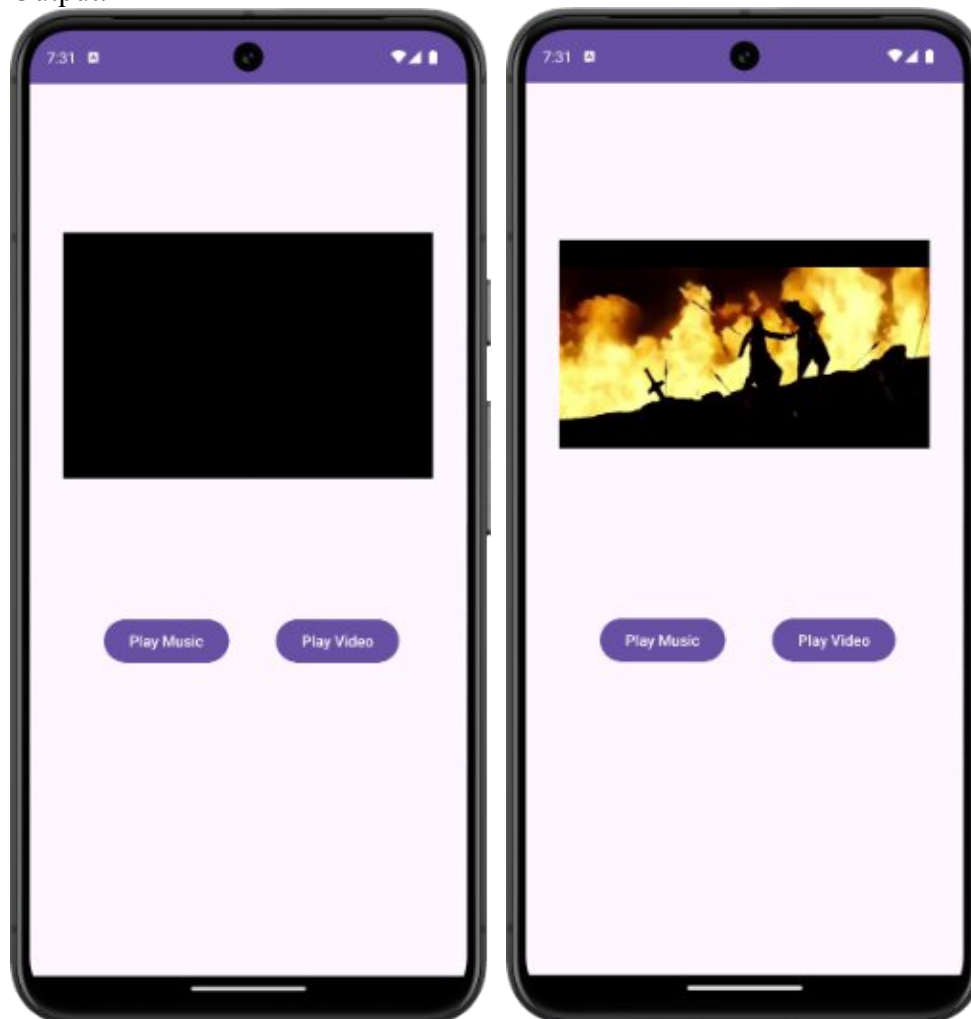
```

```
app:layout_constraintVertical_bias="0.632" />
```

```
<Button  
    android:id="@+id/video"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_marginStart="8dp"  
    android:text="Play Video"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.777"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.632" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program No: 11	Date: 29/06/2026
Program Title: Create an app that plays video from the internet.	
MainActivity.java:	
<pre> package com.example.pg11_internetvideo; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.widget.Button; import android.widget.VideoView; import android.net.Uri; import android.widget.MediaController; public class MainActivity extends AppCompatActivity { VideoView videoView; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); Button btnVideo = findViewById(R.id.play); videoView = findViewById(R.id.videoView); videoView.setZOrderOnTop(true); btnVideo.setOnClickListener(v -> { Uri uri = Uri.parse("https://www.learningcontainer.com/" + "wp-content/uploads/2020/05/" + "sample-mp4-file.mp4"); videoView.setVideoURI(uri); MediaController controller = new MediaController(MainActivity.this); videoView.setMediaController(controller); controller.setAnchorView(videoView); videoView.start(); }); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main" android:layout_width="match_parent" </pre>	

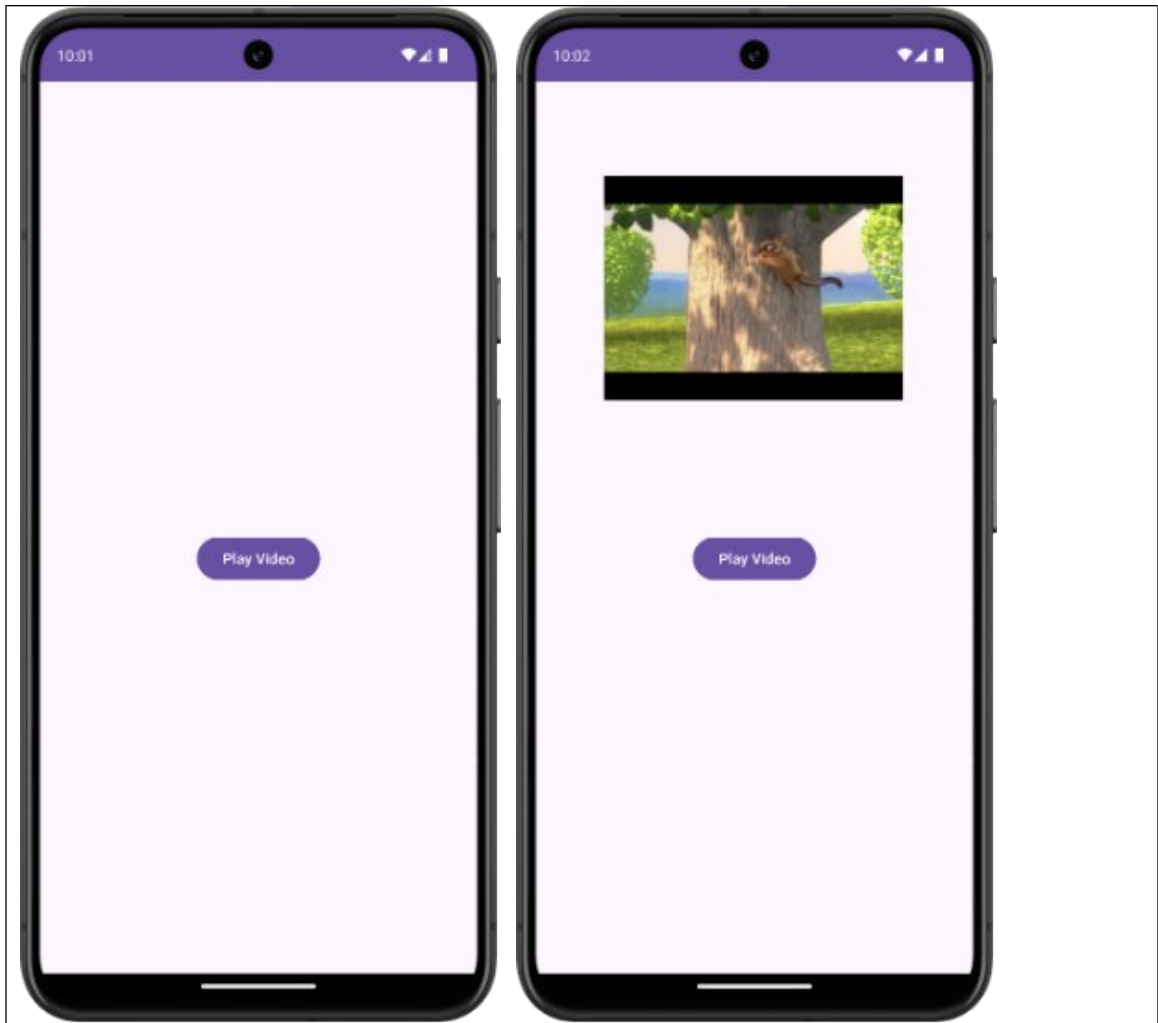
```
android:layout_height="match_parent"
tools:context=".MainActivity">
```

```
<VideoView
    android:id="@+id/videoView"
    android:layout_width="365dp"
    android:layout_height="211dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.493"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.141" />
```

```
<Button
    android:id="@+id/play"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Play Video"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.537" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



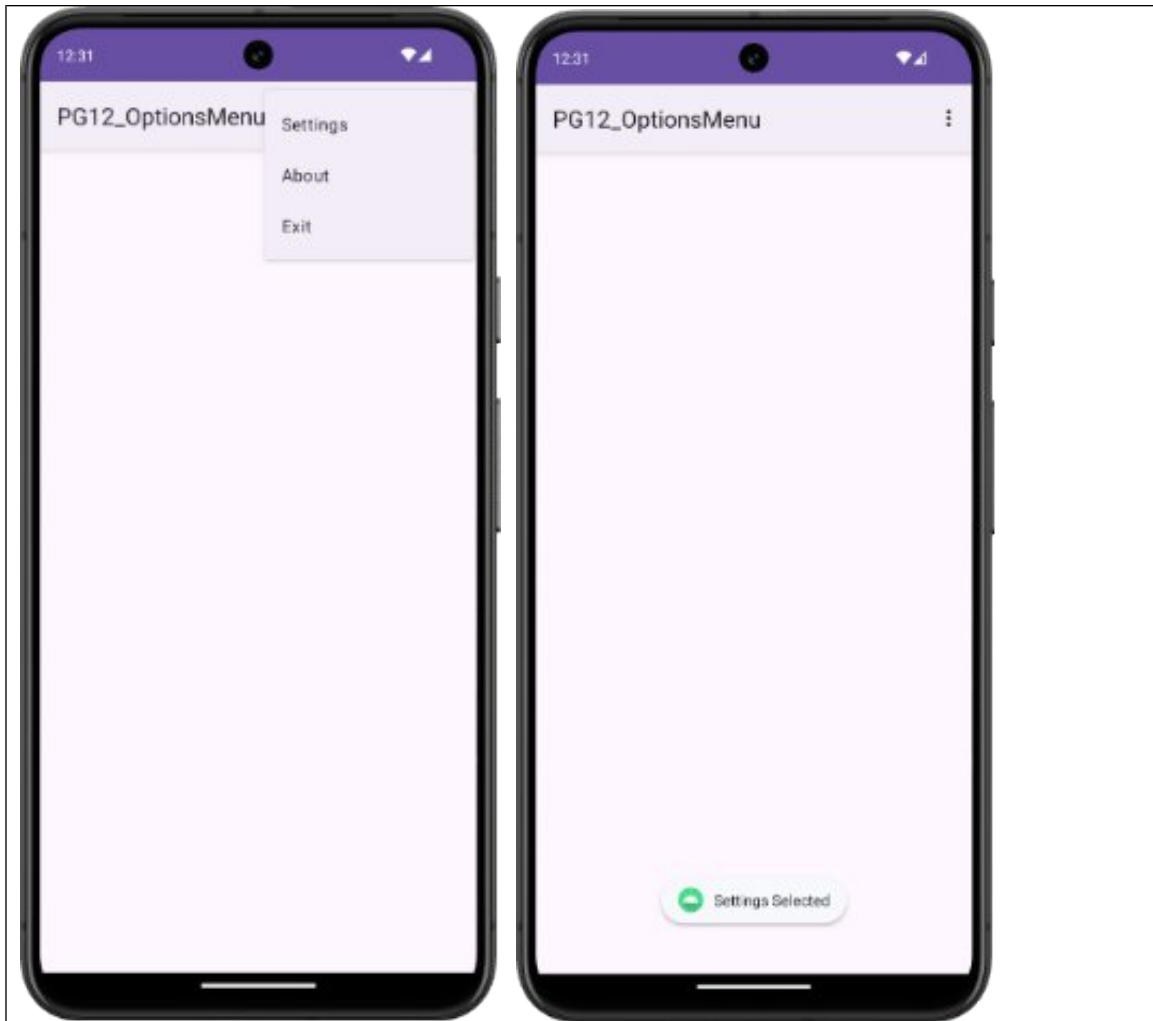
Program No: 12	Date: 29/06/2026
Program Title: Create an app to demonstrate Options Menu.	
MainActivity.java:	
<pre> package com.example.pg12_optionsmenu; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.view.Menu; import android.view.MenuItem; import android.widget.Toast; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); } @Override public boolean onCreateOptionsMenu(Menu menu) { getMenuInflater().inflate(R.menu.menu, menu); return true; } @Override public boolean onOptionsItemSelected(MenuItem item) { int id = item.getItemId(); if (id == R.id.settings) { Toast.makeText(this, "Settings Selected", Toast.LENGTH_SHORT).show(); } else if (id == R.id.about) { Toast.makeText(this, "About Selected", Toast.LENGTH_SHORT).show(); } else if (id == R.id.exit) { finish(); } return true; } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" </pre>	

```
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity"/>
```

menu.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
      xmlns:android="http://schemas.android.com/apk/res/android">
  <item
    android:id="@+id/settings"
    android:title="Settings" />
  <item
    android:id="@+id/about"
    android:title="About" />
  <item
    android:id="@+id/exit"
    android:title="Exit" />
</menu>
```

Output:



Program No: 13	Date: 29/06/2026
Program Title: Create an app that changes the background color of an activity using Context Menu.	
MainActivity.java:	
<pre> package com.example.context_menu; import android.graphics.Color; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.view.ContextMenu; import android.view.MenuInflater; import android.view.MenuItem; import android.view.View; import android.widget.TextView; public class MainActivity extends AppCompatActivity { View layout; TextView tx1; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); layout = findViewById(R.id.main); tx1 = findViewById(R.id.tx1); registerForContextMenu(tx1); } @Override public void onCreateContextMenu(ContextMenu menu, View v, ContextMenu.ContextMenuInfo menuinfo){ super.onCreateContextMenu(menu, v, menuinfo); MenuInflater inflater = getMenuInflater(); inflater.inflate(R.menu.menu_main, menu); } public boolean onOptionsItemSelected(MenuItem item) { if (item.getItemId() == R.id.red){ layout.setBackgroundColor(Color.RED); } if (item.getItemId() == R.id.blue) { layout.setBackgroundColor(Color.BLUE); } if (item.getItemId() == R.id.green) { layout.setBackgroundColor(Color.GREEN); } } </pre>	

```

        if (item.getItemId() == R.id.yellow) {
            layout.setBackgroundColor(Color.YELLOW);
        }
        return false;
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tx1"
        android:layout_width="90dp"
        android:layout_height="30dp"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.454"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.499" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

menu_main.xml:

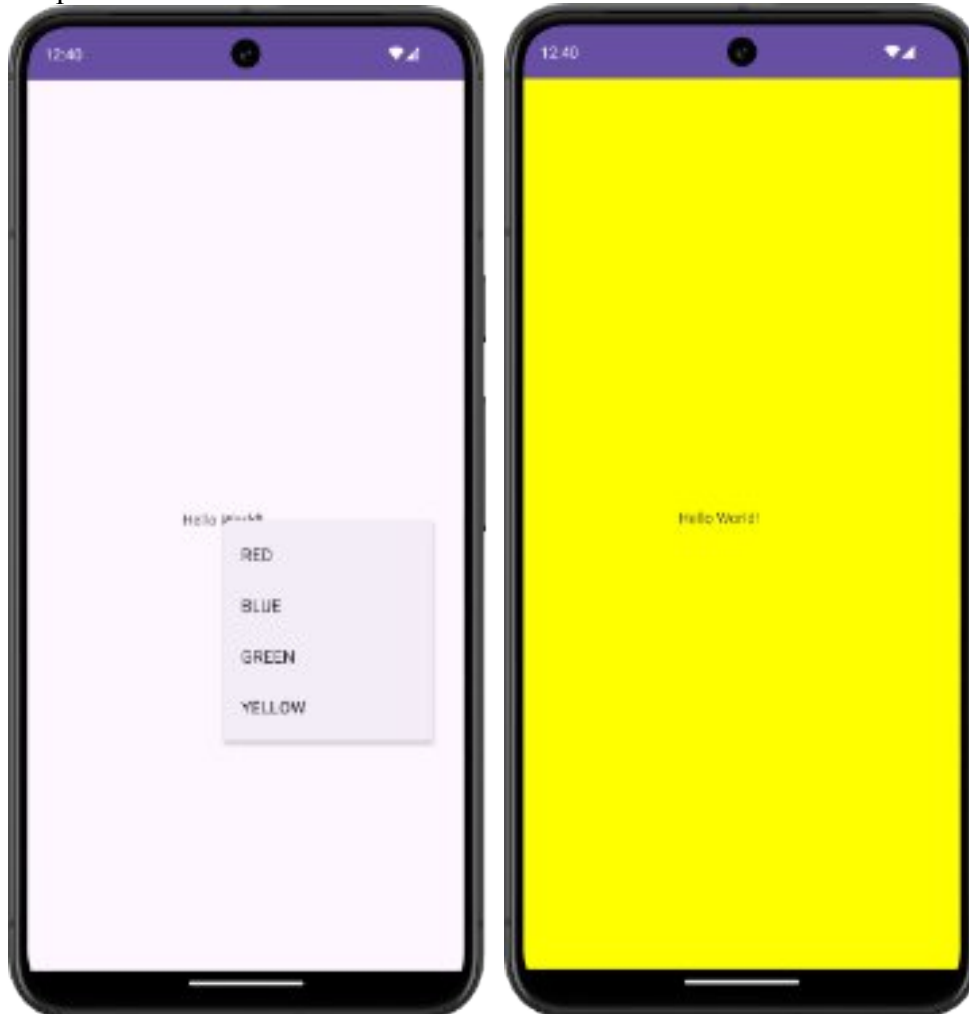
```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/red"
        android:title="RED" />
    <item
        android:id="@+id/blue"
        android:title="BLUE" />
    <item
        android:id="@+id/green"
        android:title="GREEN" />
    <item

```

```
android:id="@+id/yellow"  
android:title="YELLOW" />  
</menu>
```

Output:



Program No: 14	Date: 29/06/2026
Program Title: Create an app to demonstrate how radiobuttons and checkboxes can be used in android.	
MainActivity.java:	
<pre> package com.example.radiocheckbox; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.widget.Button; import android.widget.RadioGroup; import android.widget.RadioButton; import android.widget.CheckBox; import android.widget.Toast; public class MainActivity extends AppCompatActivity { RadioGroup gender; CheckBox ten, twelve, UG, PG; Button submit; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); gender = findViewById(R.id.gender); ten = findViewById(R.id.ten); twelve = findViewById(R.id.twelve); UG = findViewById(R.id.UG); PG = findViewById(R.id.PG); submit = findViewById(R.id.submit); submit.setOnClickListener(v ->{ int selectedId = gender.getCheckedRadioButtonId(); RadioButton rb = findViewById(selectedId); String gen = rb.getText().toString(); StringBuilder education = new StringBuilder(); if (ten.isChecked()) education.append("Tenth "); if (twelve.isChecked()) education.append("Plus Two "); if (UG.isChecked()) education.append("Undergraduate "); if (PG.isChecked()) education.append("Postgraduate "); Toast.makeText(this, gen + " " + education, Toast.LENGTH_LONG).show(); }); } </pre>	


```
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FB9E9E"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView4"
        android:layout_width="82dp"
        android:layout_height="23dp"
        android:text="Gender: "
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.124"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.114" />

    <RadioGroup
        android:id="@+id/gender"
        android:layout_width="247dp"
        android:layout_height="143dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.25"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.205">
        <RadioButton
            android:id="@+id/male"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Male" />
        <RadioButton
            android:id="@+id/female"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Female" />
    </RadioGroup>
</androidx.constraintlayout.widget.ConstraintLayout>
```

```

<RadioButton
    android:id="@+id/tg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="TG" />
</RadioGroup>

<CheckBox
    android:id="@+id/ten"
    android:layout_width="134dp"
    android:layout_height="43dp"
    android:text="X"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.155"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.527" />

<CheckBox
    android:id="@+id/UG"
    android:layout_width="133dp"
    android:layout_height="48dp"
    android:text="UG"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.71" />

<CheckBox
    android:id="@+id/twelve"
    android:layout_width="134dp"
    android:layout_height="48dp"
    android:text="XII"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.158"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.619" />

<Button
    android:id="@+id/submit"
    android:layout_width="wrap_content"

```

```

        android:layout_height="wrap_content"
        android:text="Submit"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.468"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.915" />

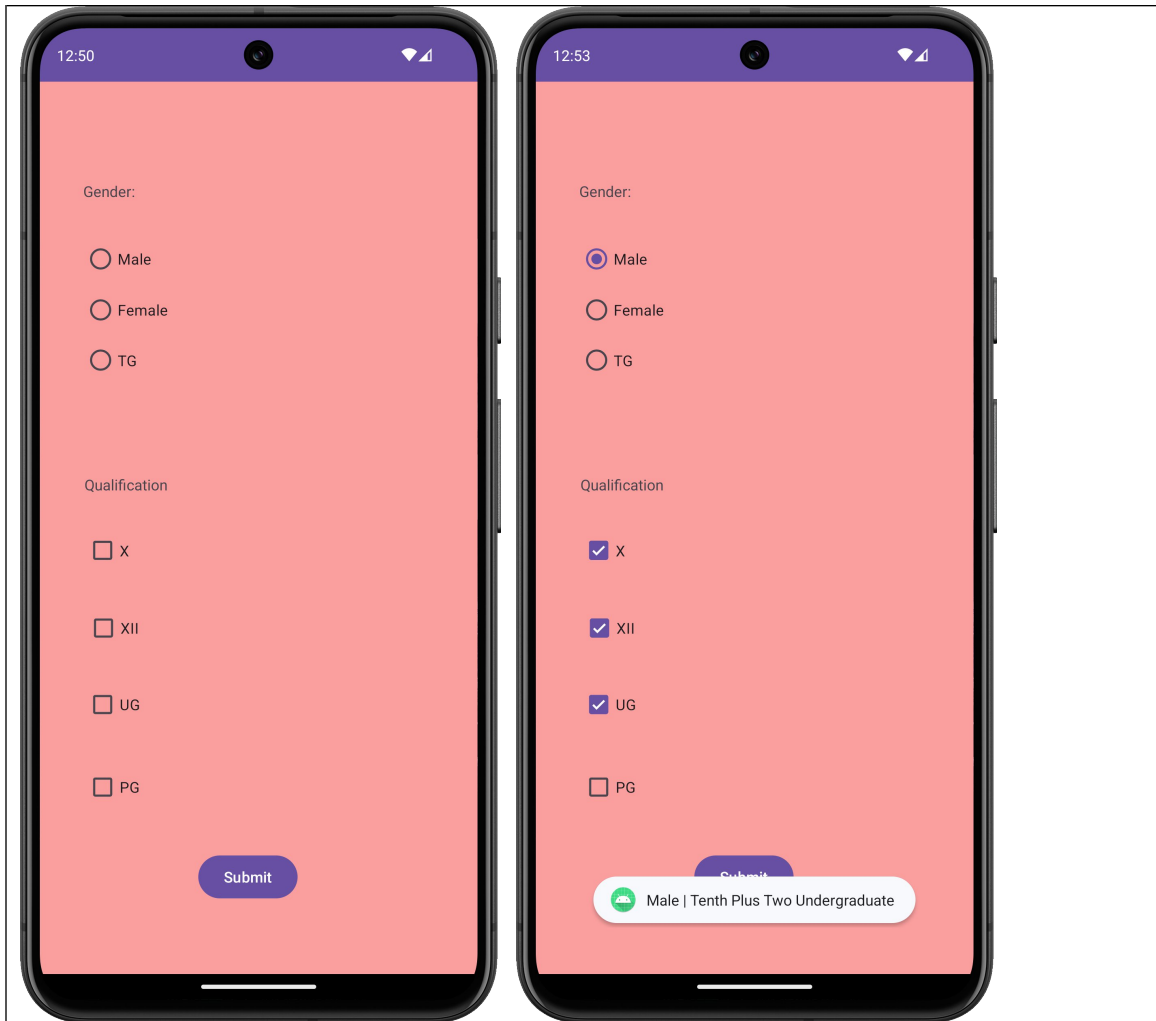
<TextView
    android:id="@+id/textView"
    android:layout_width="90dp"
    android:layout_height="22dp"
    android:text="Qualification"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.13"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.452" />

<CheckBox
    android:id="@+id/PG"
    android:layout_width="133dp"
    android:layout_height="48dp"
    android:text="PG"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.808" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 15	Date: 29/06/2026
Program Title: Create an app that launches a new activity on a button click	
MainActivity.java:	
<pre> package com.example.a2screen_welcome; import android.content.Intent; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.widget.Button; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); Button btn = findViewById(R.id.button); btn.setOnClickListener(v-> { Intent intent = new Intent(MainActivity.this, SecondaryActivity.class); startActivity(intent); }); } } </pre>	
SecondaryActivity.java:	
<pre> package com.example.a2screen_welcome; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; public class SecondaryActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_secondary); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main" android:layout_width="match_parent" android:layout_height="match_parent" tools:context=".MainActivity"> </pre>	

```

<Button
    android:id="@+id/button"
    android:layout_width="172dp"
    android:layout_height="74dp"
    android:text="Go to Next Page"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.558" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

activity_secondary.xml:

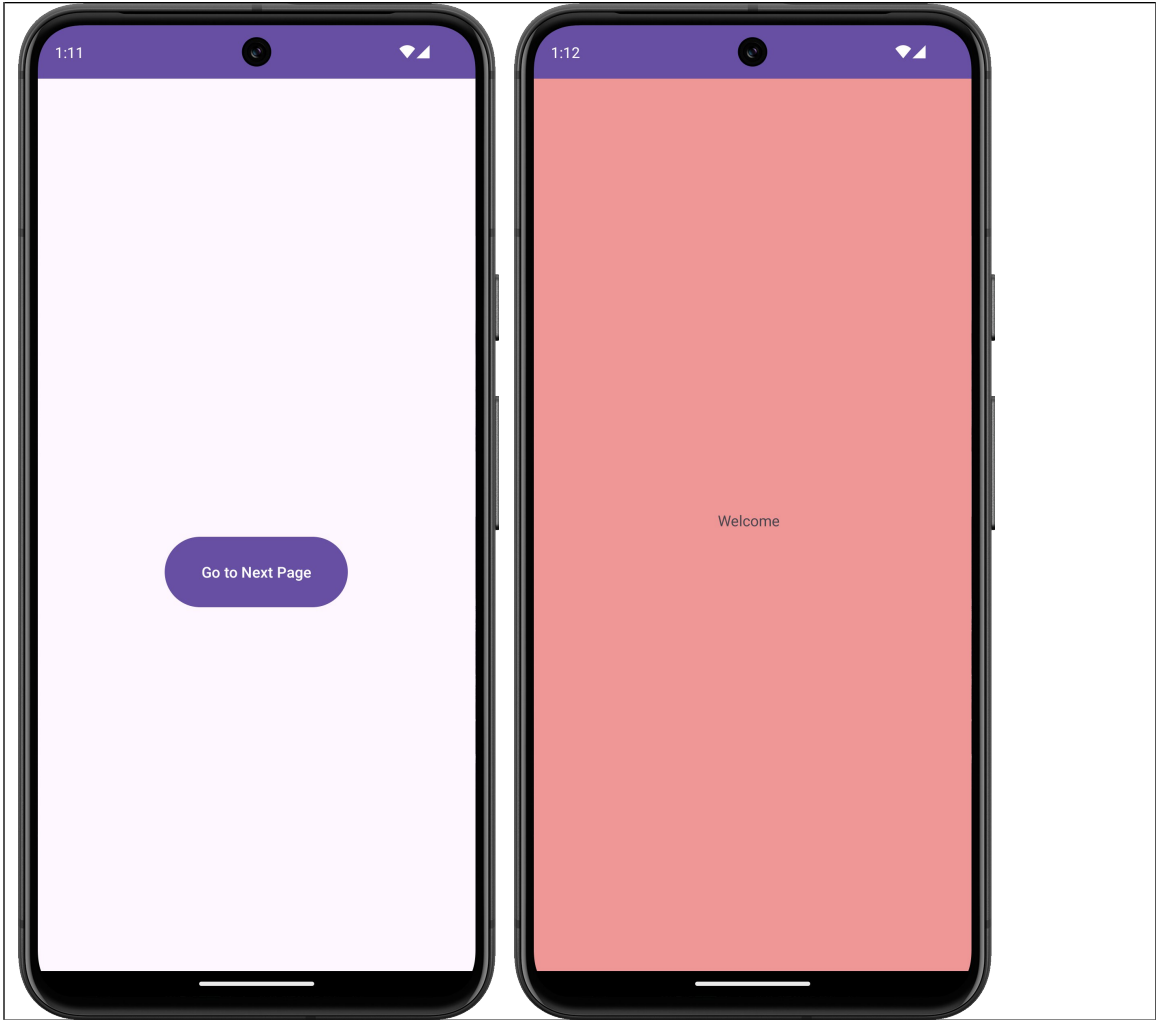
```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#EF9696"
    tools:context=".SecondaryActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="66dp"
        android:layout_height="28dp"
        android:text="Welcome"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 16	Date: 29/06/2026
Program Title: Create an app that launches a new activity on a button click to display student details with checkbox and radio buttons.	
MainActivity.java:	
<pre> package com.example.pg16_studentdetails; import android.content.Intent; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.widget.Button; import android.widget.CheckBox; import android.widget.EditText; import android.widget.RadioButton; import android.widget.RadioGroup; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); Button submit = findViewById(R.id.submit); EditText name = findViewById(R.id.name); EditText mobile = findViewById(R.id.mobile); RadioGroup department = findViewById(R.id.department); RadioGroup gender = findViewById(R.id.gender); CheckBox ai = findViewById(R.id.ai); CheckBox cloud = findViewById(R.id.cloud); CheckBox auto = findViewById(R.id.autocad); CheckBox ship = findViewById(R.id.ship); submit.setOnClickListener(v -> { int depId = department.getCheckedRadioButtonId(); RadioButton rdep = findViewById(depId); int genId = gender.getCheckedRadioButtonId(); RadioButton rgen = findViewById(genId); StringBuilder sb = new StringBuilder(); if (ai.isChecked()) sb.append("AI/ML\n"); if (cloud.isChecked()) sb.append("Cloud Computing\n"); if (auto.isChecked()) </pre>	


```

        sb.append("AutoCAD\n");

        if (ship.isChecked())
            sb.append("Ship Technology\n");

        Intent intent = new Intent(MainActivity.this, Details.class);
        intent.putExtra("name", name.getText().toString());
        intent.putExtra("mobile", mobile.getText().toString());
        intent.putExtra("courses", sb.toString());
        intent.putExtra("department", rdep.getText().toString());
        intent.putExtra("gender", rgen.getText().toString());
        startActivity(intent);
    });
}
}

```

Details.java:

```

package com.example.pg16_studentdetails;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.widget.TextView;

public class Details extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_details);

        String name = getIntent().getStringExtra("name");
        String mobile = getIntent().getStringExtra("mobile");
        String department = getIntent().getStringExtra("department");
        String gender = getIntent().getStringExtra("gender");
        String courses = getIntent().getStringExtra("courses");

        TextView details = findViewById(R.id.details);
        details.setText("Name: " + name +
            "\nMobile: " + mobile +
            "\nDepartment: " + department +
            "\nGender: " + gender +
            "\nCourses:\n" + courses);
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<TextView
    android:layout_width="68dp"
    android:layout_height="24dp"
    android:text="Name: "
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.11"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.077" />

<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Department: "
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.119"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.131" />

<RadioGroup
    android:id="@+id/department"
    android:layout_width="314dp"
    android:layout_height="134dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.391"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.207">

```

```

<RadioButton
    android:id="@+id/cs"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Computer Science"
    android:textSize="16sp" />

<RadioButton
    android:id="@+id/civil"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Civil Engineering"
    android:textSize="16sp" />

<RadioButton
    android:id="@+id/mech"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Mechanical Engineering"
    android:textSize="16sp" />
</RadioGroup>

<EditText
    android:id="@+id/name"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Name"
    android:inputType="text"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.719"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.076" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Preferred Course: "
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.123"

```

```
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.385" />
```

```
<CheckBox
    android:id="@+id/ai"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="AI/ML"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.133"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.44" />
```

```
<CheckBox
    android:id="@+id/ship"
    android:layout_width="127dp"
    android:layout_height="41dp"
    android:text="Ship Technology"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.133"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.517" />
```

```
<CheckBox
    android:id="@+id/cloud"
    android:layout_width="127dp"
    android:layout_height="41dp"
    android:text="Cloud Computing"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.693"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.44" />
```

```
<CheckBox
    android:id="@+id/autocad"
    android:layout_width="127dp"
```

```

android:layout_height="41dp"
android:text="Autocad"
android:textSize="16sp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.693"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.517" />

```

```

<TextView
    android:id="@+id/textView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Gender: "
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.107"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.585" />

```

```

<RadioGroup
    android:id="@+id/gender"
    android:layout_width="318dp"
    android:layout_height="162dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.408"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.775">

```

```

<RadioButton
    android:id="@+id/male"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Male"
    android:textSize="16sp" />

```

```

<RadioButton
    android:id="@+id/female"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Female"

```

```

        android:textSize="16sp" />

<RadioButton
    android:id="@+id/tg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="TG"
    android:textSize="16sp" />
</RadioGroup>

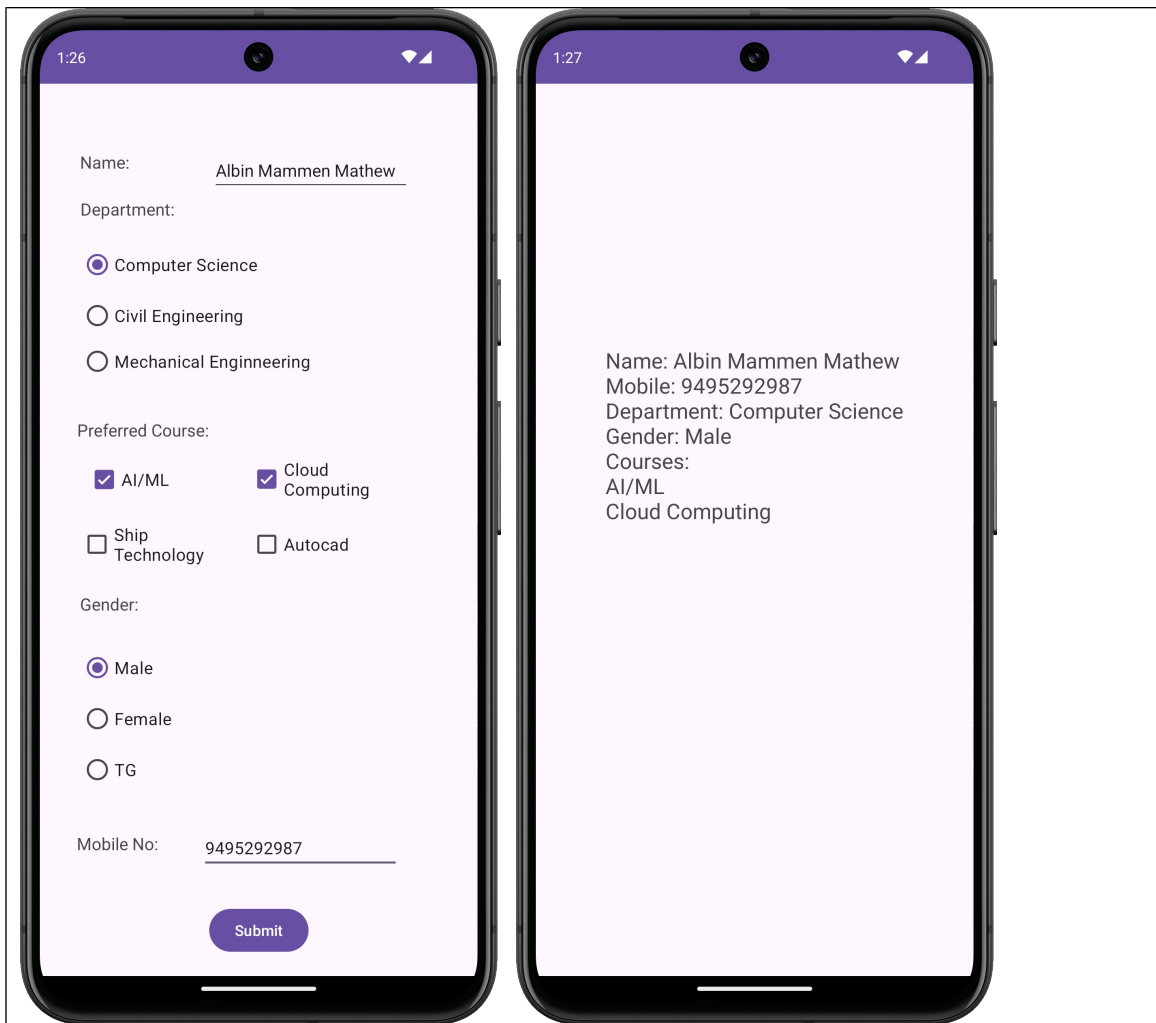
<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Mobile No: "
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.105"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.861" />

<EditText
    android:id="@+id/mobile"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="Mobile No."
    android:inputType="number"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.674"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.876" />

<Button
    android:id="@+id/submit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"

```

<pre> app:layout_constraintTop_toTopOf="parent" app:layout_constraintVertical_bias="0.976" /> </androidx.constraintlayout.widget.ConstraintLayout> </pre>
activity_details.xml: <pre> <?xml version="1.0" encoding="utf-8"?> <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main" android:layout_width="match_parent" android:layout_height="match_parent" tools:context=".Details"> <TextView android:id="@+id/details" android:layout_width="wrap_content" android:layout_height="wrap_content" android:textSize="20sp" app:layout_constraintBottom_toBottomOf="parent" app:layout_constraintEnd_toEndOf="parent" app:layout_constraintStart_toStartOf="parent" app:layout_constraintTop_toTopOf="parent" app:layout_constraintVertical_bias="0.381" /> </androidx.constraintlayout.widget.ConstraintLayout> </pre>
Output:



Program No: 17	Date: 29/06/2026
Program Title: Create an app that launches a new activity on a button click to calculate and display employee details and salary.	
MainActivity.java:	
<pre> package com.example.pg17_employeeedetails; import android.content.Intent; import android.os.Bundle; import android.view.View; import android.widget.Button; import android.widget.EditText; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { EditText name, id, sal; Button btnNext; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); name = findViewById(R.id.name); id = findViewById(R.id.id); sal = findViewById(R.id.sal); btnNext = findViewById(R.id.btnNext); btnNext.setOnClickListener(new View.OnClickListener() { @Override public void onClick(View view) { Intent intent = new Intent(MainActivity.this, SecondaryActivity.class); intent.putExtra("name", name.getText().toString()); intent.putExtra("id", id.getText().toString()); intent.putExtra("salary", sal.getText().toString()); startActivity(intent); } }); } } </pre>	
SecondaryActivity.java:	
<pre> package com.example.pg17_employeeedetails; import android.content.Intent; import android.os.Bundle; </pre>	

```

import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondaryActivity extends AppCompatActivity {
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_secondary);

        result = findViewById(R.id.result);
        Intent intent = getIntent();

        if (intent != null) {
            String name = intent.getStringExtra("name");
            String id = intent.getStringExtra("id");
            String salaryStr = intent.getStringExtra("salary");

            double basic = 0;
            try {
                if (salaryStr != null) {
                    basic = Double.parseDouble(salaryStr);
                }
            } catch (Exception e) {
                // handle parsing error
            }

            double hra = basic * 0.15;
            double da = basic * 0.10;
            double gross = basic + hra + da;

            result.setText(
                "Employee Details\n\n" +
                "Name : " + name +
                "\nEmployee ID : " + id +
                "\n\nBasic Salary : " + basic +
                "\nHRA (15%) : " + hra +
                "\nDA (10%) : " + da +
                "\nGross Salary : " + gross
            );
        }
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="match_parent">

<TextView
    android:id="@+id/tvTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Employee Details"
    android:textSize="22sp"
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.587"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.067" />

<TextView
    android:id="@+id/tvName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="20dp"
    android:layout_marginTop="20dp"
    android:text="Employee Name"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/tvTitle" />

<EditText
    android:id="@+id/name"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Enter Name"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/tvName" />

<TextView
    android:id="@+id/tvId"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="Employee ID"
    app:layout_constraintStart_toStartOf="@id/tvName"

```

```

        app:layout_constraintTop_toBottomOf="@id/name" />

<EditText
    android:id="@+id/id"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Enter ID"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/tvId" />

<TextView
    android:id="@+id/tvSalary"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="Basic Salary"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/id" />

<EditText
    android:id="@+id/sal"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:hint="Enter Salary"
    android:inputType="numberDecimal"
    app:layout_constraintStart_toStartOf="@id/tvName"
    app:layout_constraintTop_toBottomOf="@id/tvSalary" />

<Button
    android:id="@+id/btnNext"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="25dp"
    android:text="Show Details"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@id/sal" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

activity_secondary.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"

```

```

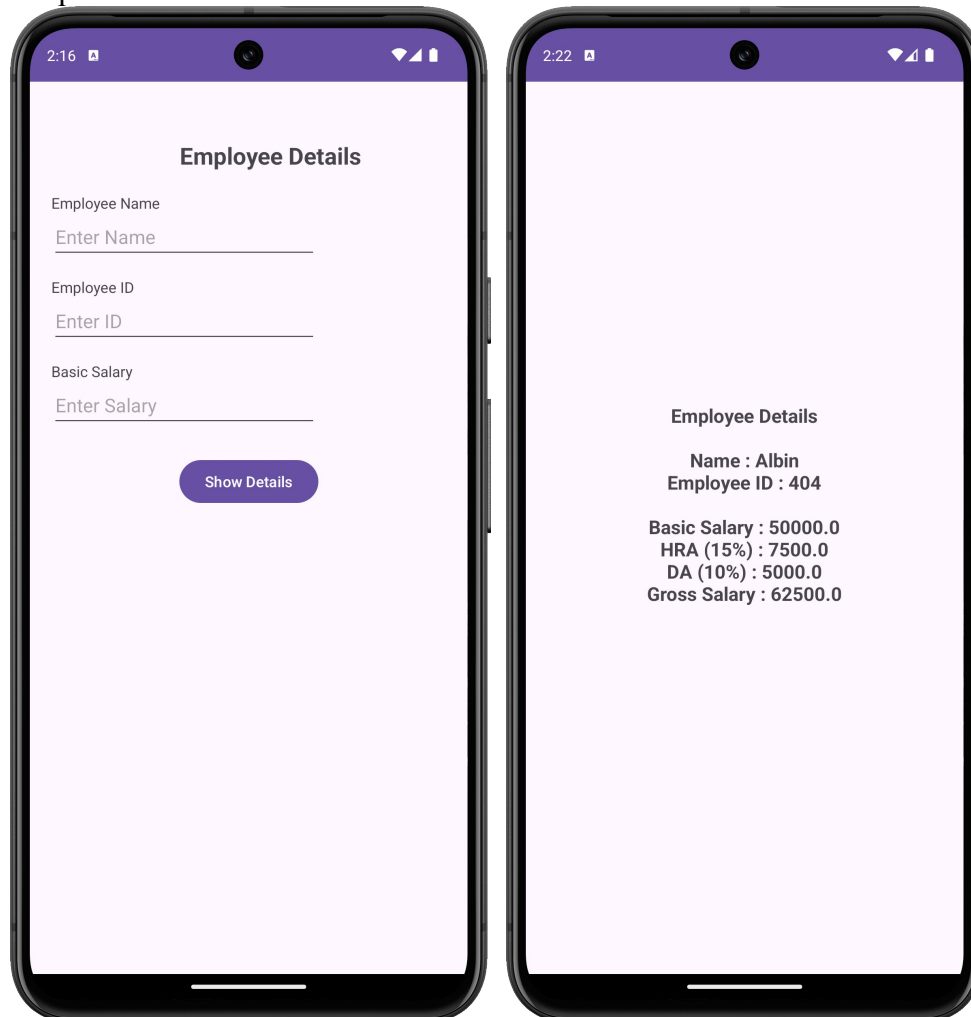
        android:layout_height="match_parent">

        <TextView
            android:id="@+id/result"
            android:layout_width="300dp"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:textSize="18sp"
            android:textStyle="bold"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toTopOf="parent" />

    </androidx.constraintlayout.widget.ConstraintLayout>

```

Output:



Program No: 18	Date: 29/06/2026
Program Title: Create an app to broadcast a custom intent.	
MainActivity.java:	
<pre> package com.example.broadcast; import android.content.Intent; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; import android.widget.Button; public class MainActivity extends AppCompatActivity { public static final String MY_ACTION = "com.example.broadcast.MY_ACTION"; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); Button send = findViewById(R.id.send); send.setOnClickListener(v -> { Intent intent = new Intent(MY_ACTION); intent.putExtra("message", "Hello from MainActivity!!"); intent.setPackage(getPackageName()); sendBroadcast(intent); }); } } </pre>	
MyReceiver.java:	
<pre> package com.example.broadcast; import android.content.BroadcastReceiver; import android.content.Context; import android.content.Intent; import android.widget.Toast; public class MyReceiver extends BroadcastReceiver { @Override public void onReceive(Context context, Intent intent) { String message = intent.getStringExtra("message"); Toast.makeText(context, "Custom Broadcast Received: " + message, Toast.LENGTH_SHORT).show(); } } </pre>	

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/send"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.52"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.614" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

AndroidManifest.xml:

```
<receiver
    android:name=".MyReceiver"
    android:enabled="true"
    android:exported="true">
    <intent-filter>
        <action android:name="com.example.broadcast.MY_ACTION" />
    </intent-filter>
</receiver>
```

Output:



Program No: 19	Date: 29/06/2026
Program Title: Create an app to broadcast a system intent (AIRPLANE MODE CHANGED).	
MainActivity.java:	
<pre> package com.example.pg19_airplanemode; import android.content.Intent; import android.content.IntentFilter; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { private final AirplaneModeReceiver receiver = new AirplaneModeReceiver(); @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); IntentFilter filter = new IntentFilter(Intent.ACTION_AIRPLANE_MODE_CHANGED); registerReceiver(receiver, filter); } @Override protected void onDestroy() { super.onDestroy(); unregisterReceiver(receiver); } } </pre>	
AirplaneModeReceiver.java:	
<pre> package com.example.pg19_airplanemode; import android.content.BroadcastReceiver; import android.content.Context; import android.content.Intent; import android.widget.Toast; public class AirplaneModeReceiver extends BroadcastReceiver { @Override public void onReceive(Context context, Intent intent) { if (intent.ACTION_AIRPLANE_MODE_CHANGED.equals(intent.getAction())) { boolean isOn = intent.getBooleanExtra("state", false); String msg = isOn ? "Airplane Mode turned ON" </pre>	

```

        : "Airplane Mode turned OFF";
        Toast.makeText(context, msg, Toast.LENGTH_SHORT).show();
    }
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" />

```

AndroidManifest.xml:

```

<receiver
    android:name=".AirplaneModeReceiver"
    android:enabled="true"
    android:exported="true">
</receiver>

```

Output:



Program No: 20	Date: 29/06/2026
Program Title: Create an app that launches a new activity on a button click.	
MainActivity.java:	
<pre> package com.example.batterylow; import android.content.Intent; import android.content.IntentFilter; import android.os.Bundle; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { PowerReceiver receiver = new PowerReceiver(); @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); IntentFilter filter = new IntentFilter(Intent.ACTION_BATTERY_CHANGED); registerReceiver(receiver, filter); } @Override protected void onDestroy() { super.onDestroy(); unregisterReceiver(receiver); } } </pre>	
PowerReceiver.java:	
<pre> package com.example.batterylow; import android.content.BroadcastReceiver; import android.content.Context; import android.content.Intent; import android.os.BatteryManager; import android.widget.Toast; public class PowerReceiver extends BroadcastReceiver { @Override public void onReceive(Context context, Intent intent) { int level = intent.getIntExtra(BatteryManager.EXTRA_LEVEL, 0); int scale = intent.getIntExtra(BatteryManager.EXTRA_SCALE, 100); int percent = level * 100 / scale; if (percent <= 15) { </pre>	

```
        Toast.makeText(context, "Battery Low: " + percent + "%",
            Toast.LENGTH_SHORT).show();
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" />
```

AndroidManifest.xml:

```
<receiver
    android:name=".PowerReceiver"
    android:enabled="true"
    android:exported="true">
</receiver>
```

Output:



Program No: 21	Date: 29/06/2026
Program Title: Create an alarm application.	
MainActivity.java:	
<pre> package com.example.pg20_alarmapp; import android.app.AlarmManager; import android.app.PendingIntent; import android.content.Intent; import android.os.Bundle; import android.widget.Button; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { Button btnAlarm; @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); btnAlarm = findViewById(R.id.btnAlarm); btnAlarm.setOnClickListener(v -> { AlarmManager alarmManager = (AlarmManager) getSystemService(ALARM_SERVICE); Intent intent = new Intent(MainActivity.this, AlarmReceiver.class); PendingIntent pendingIntent = PendingIntent.getBroadcast(this, 0, intent, PendingIntent.FLAG_IMMUTABLE); long time = System.currentTimeMillis() + 10000; // 10 seconds alarmManager.set(AlarmManager.RTC_WAKEUP, time, pendingIntent); }); } } </pre>	
AlarmReceiver.java:	
<pre> package com.example.pg20_alarmapp; import android.content.BroadcastReceiver; import android.content.Context; import android.content.Intent; import android.widget.Toast; public class AlarmReceiver extends BroadcastReceiver { @Override public void onReceive(Context context, Intent intent) { </pre>	

```
        Toast.makeText(context, "Alarm Triggered!", Toast.LENGTH_LONG).show();
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

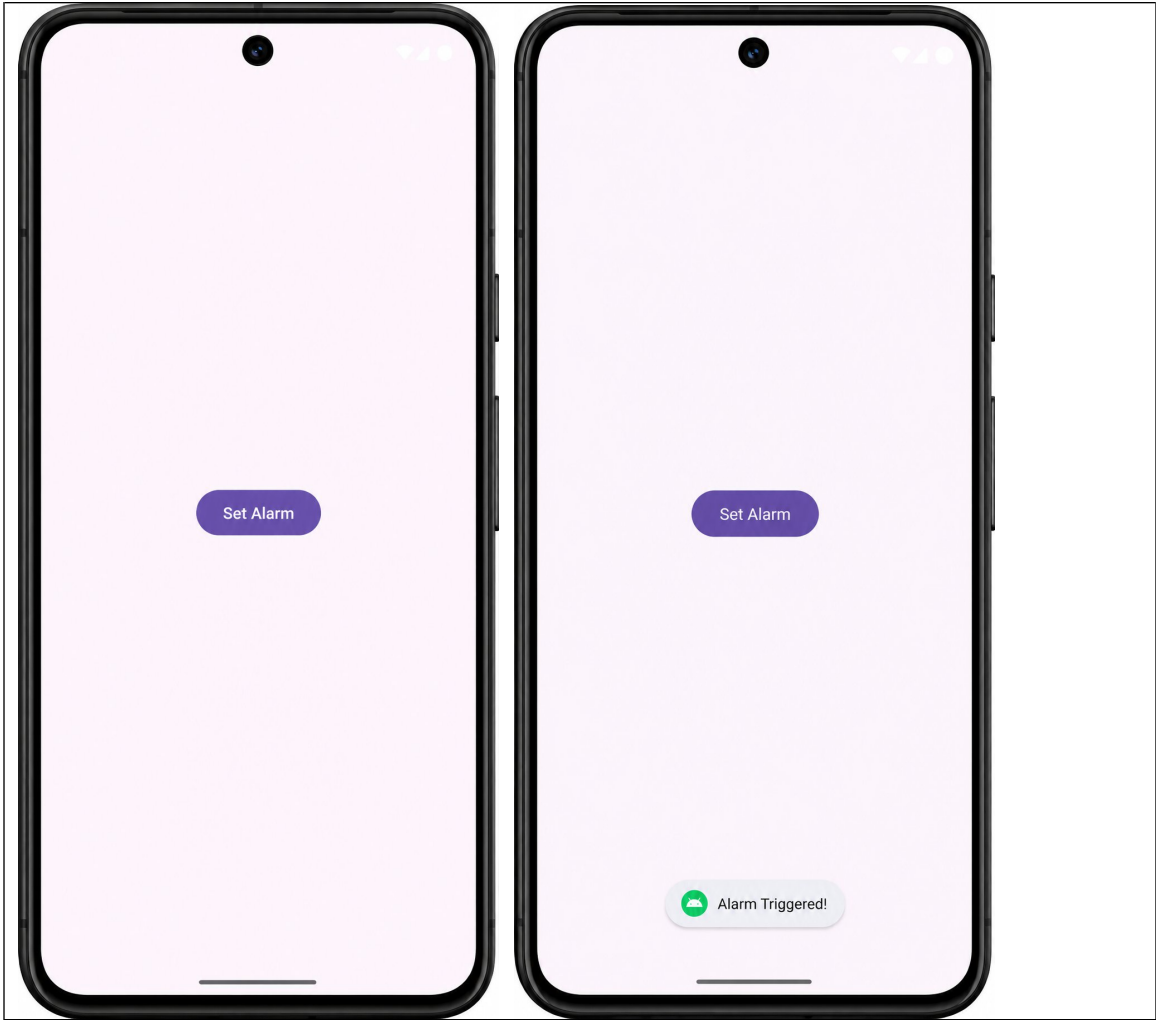
    <Button
        android:id="@+id/btnAlarm"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Set Alarm" />

</LinearLayout>
```

AndroidManifest.xml:

```
<receiver
    android:name=".AlarmReceiver"
    android:enabled="true"
    android:exported="false" />
```

Output:



Program No: 22	Date: 29/06/2026
Program Title: Create an app that displays a simple ListView. (Contacts list).	
MainActivity.java:	
<pre> package com.example.contactlist; import android.os.Bundle; import android.widget.ArrayAdapter; import android.widget.ListView; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { @Override protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState); setContentView(R.layout.activity_main); String[] contacts = { "Albin - 9876543210", "Shyam - 9123456780", "Jubin - 9988776655", "Sharvin - 9876501234", "Anandhu - 9012345678" }; ListView listView = findViewById(R.id.listview); ArrayAdapter<String> adapter = new ArrayAdapter<>(this, android.R.layout.simple_list_item_1, contacts); listView.setAdapter(adapter); } } </pre>	
activity_main.xml:	
<pre> <?xml version="1.0" encoding="utf-8"?> <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" android:layout_width="match_parent" android:layout_height="match_parent" android:orientation="vertical" android:padding="16dp"> <TextView android:layout_width="wrap_content" android:layout_height="wrap_content" android:layout_gravity="center" android:layout_marginTop="40dp" </pre>	

```
android:text="Contact List"  
android:textSize="24sp" />
```

```
<ListView  
  android:id="@+id/listview"  
  android:layout_width="match_parent"  
  android:layout_height="0dp"  
  android:layout_weight="1" />
```

```
</LinearLayout>
```

Output:

